Pattikonda Surendra

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Education

RGM College Of Engineering And Technology, Nandyal, Andhra Pradesh — *oct 2022 – Current*
B.tech in Computer Science (Specialization: Data Science) — GPA: 8.23/10
Relevant Coursework: Data Structures, Machine Learning, AI, Computer Vision, Big Data

MJPAPBCWR Junior College — *sep 2020 – May 2022*
Intermediate (MPC) — Percentage:89

Skills

Programming Languages: Python, C++, JavaScript, C
Web Technologies: HTML, CSS, React.js, Node.js, Express.js
Data Science & ML: Pandas, NumPy, scikit-learn, Matplotlib, TensorFlow, NLTK
Tools & Platforms: VS Code, Jupyter Notebook, Google Colab
Other Skills: Data Structures & Algorithms, OOPs, DBMS, OS, Computer Networks

Projects

MCQ Wizard – AI-Powered MCQ Generator — Live Demo

- Developed a full-stack web app that converts PDF/DOCX study materials into MCQs using Google Gemini AI, reducing manual effort by over 80%.
- Built a React + Tailwind frontend for file upload, difficulty selection, live MCQ preview, and downloadable output.
- Designed a FastAPI backend with CORS support, integrated Gemini API for NLP-based MCQ generation using Google LLM.
- Enabled export of generated MCQs in PDF (jsPDF), DOCX (html-docx-js), and TXT formats for educational use.
- Planned enhancements like user authentication, multilingual support, and topic-wise generation.
- **Technologies:** React.js, Tailwind CSS, FastAPI, Google Gemini API, jsPDF, html-docx-js

Smart Agriculture Assistant — Live Demo

- Built a smart assistant to detect plant diseases from images using CNN models.
- Suggested appropriate pesticides and nearby agricultural stores based on disease detection results.
- Incorporated weather-based pesticide spraying recommendations using OpenWeather API.
- Developed the frontend using React and connected it to a FastAPI backend for seamless data flow.
- **Skills:** Machine Learning, Computer Vision, CNN, TensorFlow, React.js, FastAPI, REST APIs, OpenWeather API

Achievements

- Ranked **1st** in multiple inter-departmental coding competitions at RGMCET (2023, 2024), outperforming 200+ peers.
- Solved over **300+ coding problems** across platforms including **LeetCode** and **GeeksforGeeks**, enhancing DSA proficiency.

Leadership & Activities

- **Lead**, CODES Club (Data Science Division), RGMCET — organized workshops, tech talks, and peer-learning sessions.
- Mentored 60+ junior students in data structures, algorithmic thinking, and problem-solving via hands-on sessions.
- Led and facilitated college-level hackathons and coding bootcamps.

Certificates

- **Supervised Machine Learning: Regression and Classification** – DeepLearning.AI & Stanford University (Coursera)
  Mastered linear/logistic regression, gradient descent, and model evaluation techniques.
- **Getting Started with Deep Learning** – NVIDIA
  Gained hands-on experience with neural networks, activation functions, and training workflows.
- **Front-End Web Development Certification** – edX
  Developed proficiency in HTML, CSS, JavaScript, and responsive UI design using modern web technologies.